Inventory and Stock Management System (ISMS) – Work Breakdown Structure and Project Schedule

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Aim

To plan the Inventory and Stock Management System (ISMS) project by preparing a Work Breakdown Structure (WBS) and a Project Schedule (Gantt chart with activity list, durations and dependencies), starting on 28 July 2026 and based on the seven core features and non-functional requirements in the team's Software Requirements Specification (SRS).

- **Work Breakdown Structure** – divides the project into major phases and work packages so that every deliverable is identified, numbered and assignable.
- **Project Schedule** – sequences those work packages with durations, predecessors, start and finish dates, and the total project duration.

Work Breakdown Structure

The WBS divides the complete ISMS project (1.0) into seven major phases and 34 work packages. Every element carries a hierarchical number, so each package can be scheduled, assigned and tracked on its own.

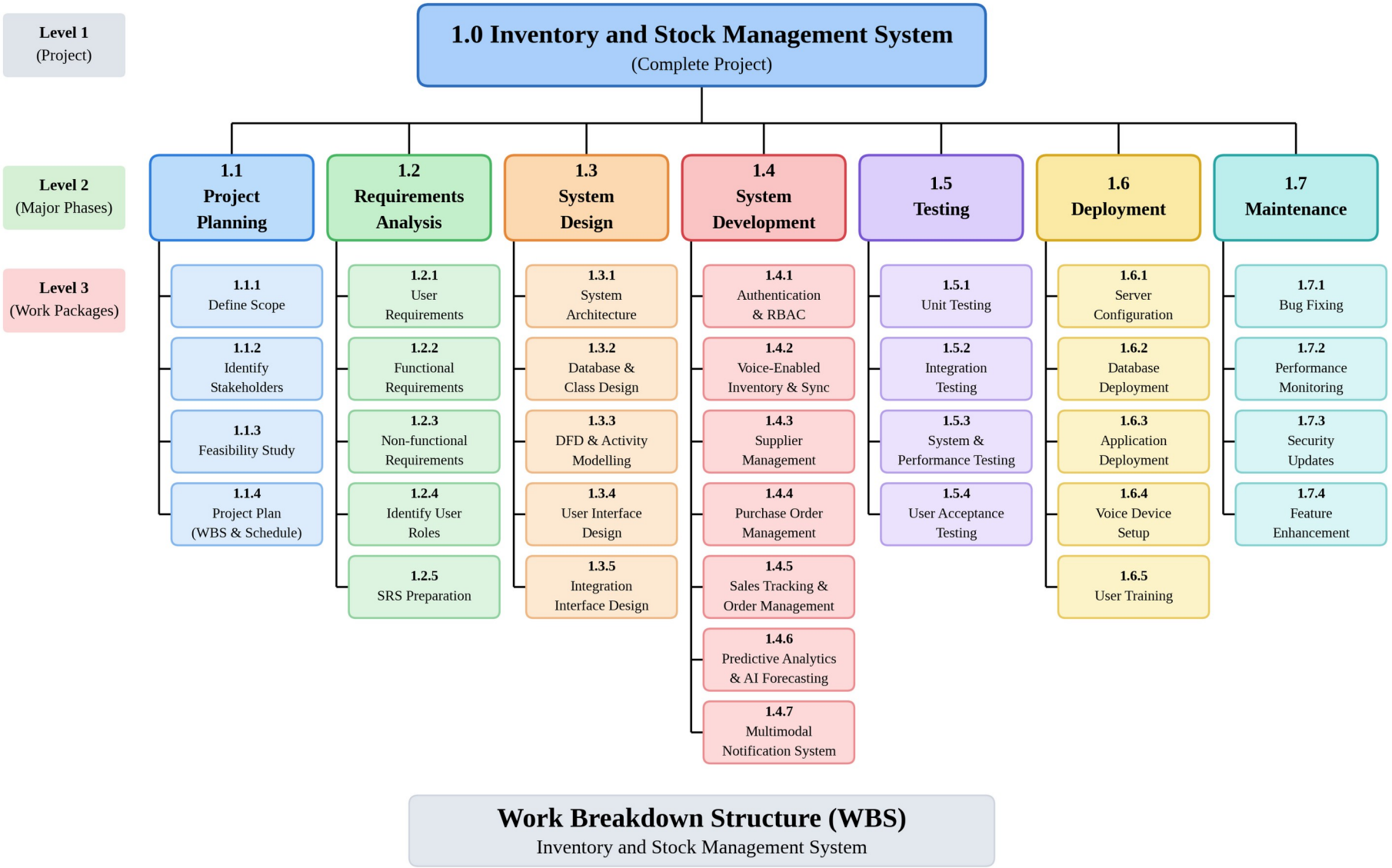


Figure 1: Work Breakdown Structure – Inventory and Stock Management System

- **Level 1 – Project:** 1.0 Inventory and Stock Management System, the complete deliverable.
- **Level 2 – Major phases:** 1.1 Project Planning, 1.2 Requirements Analysis, 1.3 System Design, 1.4 System Development, 1.5 Testing, 1.6 Deployment and 1.7 Maintenance.
- **Level 3 – Work packages:** the smallest units of work, such as 1.4.2 Voice-Enabled Inventory & Sync or 1.5.4 User Acceptance Testing.

System Development is broken down exactly as the SRS is: its seven work packages are the seven system features of SRS Section 4. The other phases cover the SRS's supporting requirements, as the table shows.

Work package	SRS reference
1.4.1 Authentication & RBAC	4.1 Authentication & Role-Based Access Control (High)
1.4.2 Voice-Enabled Inventory & Sync	4.2 Voice-Enabled Inventory & Omnichannel Sync (High)
1.4.3 Supplier Management	4.3 Supplier Management (Medium)
1.4.4 Purchase Order Management	4.4 Purchase Order (PO) Management (High)
1.4.5 Sales Tracking & Order Management	4.5 Sales Tracking & Order Management (High)
1.4.6 Predictive Analytics & AI Forecasting	4.6 Predictive Analytics & AI Forecasting (Medium)
1.4.7 Multimodal Notification System	4.7 Multimodal Notification System (Medium)
1.2.3 Non-functional Requirements, 1.5.3 System & Performance Testing	5.1–5.4 Performance (PE-1 to PE-4), security (SE-1 to SE-5), availability, scalability
1.3.5 Integration Interface Design	3.3 Software Interfaces: marketplace webhooks, speech-to-text / text-to-speech, ML forecasting, payment gateway, e-mail / SMS
1.6.4 Voice Device Setup	OE-3: Android handhelds or wearables with microphone and speaker
1.6.5 User Training	UD-2: role-based onboarding walkthrough
1.7.4 Feature Enhancement	1.4 and 6: later releases (more voice languages, more marketplaces)

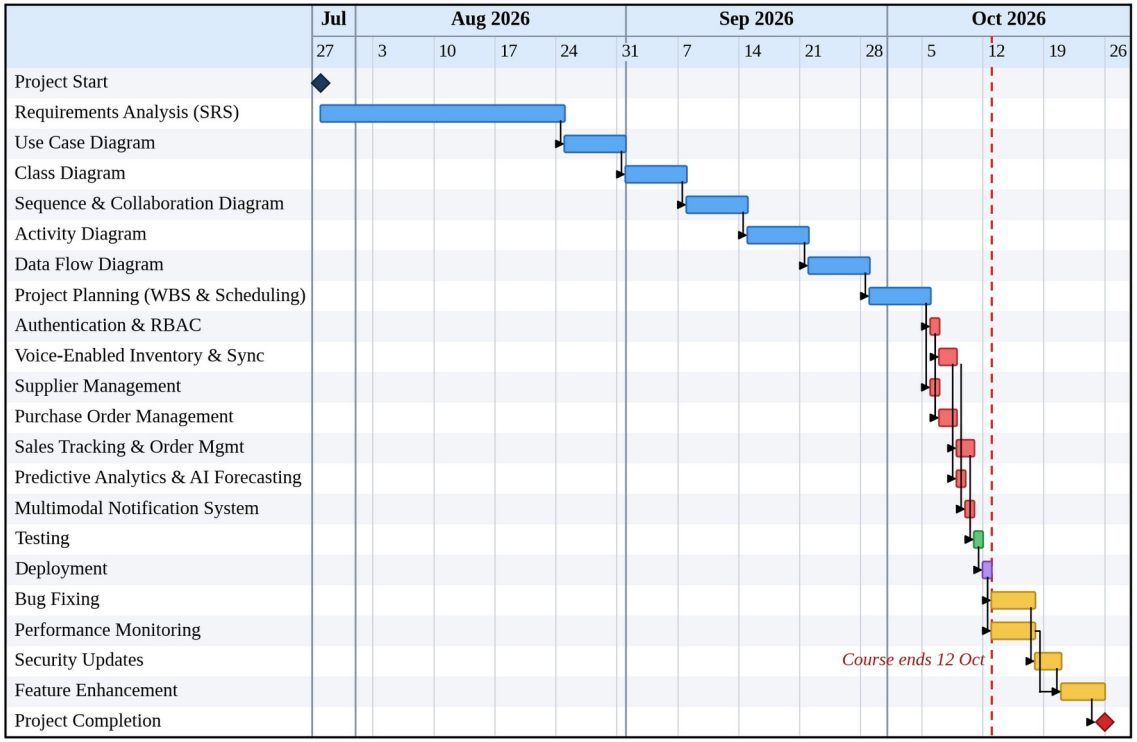
Project Schedule

The project runs from 28 July 2026 to 25 October 2026, 90 days in all, and is fitted to the Software Engineering course timeline (28 July to 12 October 2026): the course work is complete on 12 October 2026 (77 days) and Maintenance follows after the course. Activity IDs match the WBS, so every row of the chart can be traced to a work package.

Project Schedule – Inventory and Stock Management System

(Gantt Chart with Activity List, Duration and Dependencies)

ID	Activity	Duration (days)	Predecessor(s)	Start Date	Finish Date
1.0	Project Start	0	–	28 Jul 2026	28 Jul 2026
1.2	Requirements Analysis (SRS)	28	–	28 Jul 2026	24 Aug 2026
1.3a	Use Case Diagram	7	1.2	25 Aug 2026	31 Aug 2026
1.3b	Class Diagram	7	1.3a	01 Sep 2026	07 Sep 2026
1.3c	Sequence & Collaboration Diagram	7	1.3b	08 Sep 2026	14 Sep 2026
1.3d	Activity Diagram	7	1.3c	15 Sep 2026	21 Sep 2026
1.3e	Data Flow Diagram	7	1.3d	22 Sep 2026	28 Sep 2026
1.1	Project Planning (WBS & Scheduling)	7	1.3e	29 Sep 2026	05 Oct 2026
1.4.1	Authentication & RBAC	1	1.1	06 Oct 2026	06 Oct 2026
1.4.2	Voice-Enabled Inventory & Sync	2	1.4.1	07 Oct 2026	08 Oct 2026
1.4.3	Supplier Management	1	1.1	06 Oct 2026	06 Oct 2026
1.4.4	Purchase Order Management	2	1.4.1, 1.4.3	07 Oct 2026	08 Oct 2026
1.4.5	Sales Tracking & Order Mgmt	2	1.4.2	09 Oct 2026	10 Oct 2026
1.4.6	Predictive Analytics & AI Forecasting	1	1.4.4	09 Oct 2026	09 Oct 2026
1.4.7	Multimodal Notification System	1	1.4.2, 1.4.4, 1.4.6	10 Oct 2026	10 Oct 2026
1.5	Testing	1	1.4.5, 1.4.7	11 Oct 2026	11 Oct 2026
1.6	Deployment	1	1.5	12 Oct 2026	12 Oct 2026
1.7.1	Bug Fixing	5	1.6	13 Oct 2026	17 Oct 2026
1.7.2	Performance Monitoring	5	1.6	13 Oct 2026	17 Oct 2026
1.7.3	Security Updates	3	1.7.1	18 Oct 2026	20 Oct 2026
1.7.4	Feature Enhancement	5	1.7.2, 1.7.3	21 Oct 2026	25 Oct 2026
1.0	Project Completion	0	1.7.4	25 Oct 2026	25 Oct 2026



Legend: Planning / Analysis / Design · Development · Testing · Deployment · Maintenance · Dependency · Milestone

Total Project Duration: 90 days
(28 Jul 2026 – 25 Oct 2026)
Course period: 77 days (28 Jul – 12 Oct); maintenance follows

Figure 2: Project Schedule – Inventory and Stock Management System (Gantt chart)

The schedule is built on these assumptions:

- Durations are in calendar days; every dependency is finish-to-start, so an activity begins the day after its last predecessor finishes.
- Until 5 October 2026 the course work is the set of project diagrams. Each of the six diagrams (Use Case, Class, Sequence & Collaboration, Activity, Data Flow, and WBS & Scheduling) is given one week, and the time left over, four weeks, goes to the SRS (1.2), which starts on 28 July.
- System Design (1.3) is delivered through the five diagrams 1.3a to 1.3e, and Project Planning (1.1) through the WBS & Scheduling diagram, which is due on 5 October 2026.
- The last week of the course holds System Development (6 to 10 October, 5 days), Testing (11 October, 1 day) and Deployment (12 October, 1 day). Testing and Deployment are one-day activities carried out by both team members, covering work packages 1.5.1–1.5.4 and 1.6.1–1.6.5.
- The team has two members, so no more than two development packages run in parallel; 1.4.7 therefore follows 1.4.6, which the same member builds.
- Maintenance (1.7) starts after the course ends, on 13 October 2026.

Phase / diagram	**Start**	**Finish**	**Days**
Requirements Analysis – SRS (1.2)	28 Jul 2026	24 Aug 2026	28
Use Case Diagram (1.3a)	25 Aug 2026	31 Aug 2026	7
Class Diagram (1.3b)	1 Sep 2026	7 Sep 2026	7
Sequence & Collaboration Diagram (1.3c)	8 Sep 2026	14 Sep 2026	7
Activity Diagram (1.3d)	15 Sep 2026	21 Sep 2026	7
Data Flow Diagram (1.3e)	22 Sep 2026	28 Sep 2026	7
WBS & Scheduling Diagram – Project Planning (1.1)	29 Sep 2026	5 Oct 2026	7
System Development (1.4)	6 Oct 2026	10 Oct 2026	5
Testing (1.5)	11 Oct 2026	11 Oct 2026	1
Deployment (1.6)	12 Oct 2026	12 Oct 2026	1
Maintenance (1.7)	13 Oct 2026	25 Oct 2026	13

The order of the development packages follows the dependencies between the SRS features:

- Authentication & RBAC (1.4.1) and Supplier Management (1.4.3) start together on 6 October, one day each. Every module is role-restricted, so Authentication comes first.
- Voice-Enabled Inventory & Sync (1.4.2) follows Authentication, and Purchase Order Management (1.4.4) follows Authentication and Supplier Management; both take two days (7–8 October).
- Sales Tracking (1.4.5), two days, needs the stock model from 1.4.2 to deduct stock. Predictive Analytics (1.4.6) follows PO Management, and Multimodal Notifications (1.4.7) need low-stock events from 1.4.2 and overdue-PO events from 1.4.4.
- Every development package lasts one or two days, that is 8 to 16 hours of work, which keeps each task within the 8/80 rule.

The critical path, which has no slack, is 1.2 → 1.3a → 1.3b → 1.3c → 1.3d → 1.3e → 1.1 → 1.4.1 → 1.4.2 → 1.4.5 → 1.5 → 1.6 → 1.7.1 → 1.7.3 → 1.7.4. The chain 1.4.3 → 1.4.4 → 1.4.6 → 1.4.7

also has no slack, because the team has only two members. Only Performance Monitoring (1.7.2) has slack, three days.

Conclusion

The Work Breakdown Structure and Project Schedule for the Inventory and Stock Management System were prepared from the team's SRS. The WBS splits the project into seven phases and 34 work packages, with the seven system features of SRS Section 4 forming the development packages and the SRS's interface, performance, deployment and documentation requirements covered by packages in the other phases.

The schedule sequences those packages with explicit durations and finish-to-start dependencies, and fits them to the course timeline. Starting on 28 July 2026, the SRS takes four weeks (to 24 August), each of the five design diagrams and the WBS & Scheduling diagram takes one week (to 5 October), and development, testing and deployment are completed between 6 and 12 October 2026. Maintenance follows after the course and ends on 25 October 2026, 90 days after the start. The critical path runs through the SRS, all six diagrams, the development of the voice-enabled inventory and sales tracking features, testing and deployment. Together, the two diagrams give the team a clear basis for assigning work, tracking progress and spotting delays early.